

Disproof of Conjecture 3490: chromatic polynomials determine chromatic numbers

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Claim. Conjecture 3490 asserts the existence of graph families with identical chromatic polynomial and spectral radius but different chromatic numbers, with minimum order ten. We prove that even a single pair of finite graphs with identical chromatic polynomial and different chromatic numbers cannot exist. Thus the extra spectral-radius condition and the stated minimum order cannot repair the assertion.

Definitions. For a finite graph G and a nonnegative integer k , a proper k -coloring is a function $V(G) \rightarrow \{0, \dots, k-1\}$ assigning different colors to adjacent vertices. The chromatic polynomial P_G is characterized by

$$P_G(k) = \#\{\text{proper } k\text{-colorings of } G\} \quad (k \in \mathbb{N}).$$

The chromatic number $\chi(G)$ is the least k admitting a proper k -coloring. These conventions also cover the empty graph, whose chromatic number is zero.

Proof. For every $k \in \mathbb{N}$,

$$G \text{ is } k\text{-colorable} \iff P_G(k) > 0.$$

Consequently $P_G = P_H$ implies that G and H admit proper k -colorings for exactly the same values of k . Their least such values agree, so $\chi(G) = \chi(H)$. Equivalently, if $\chi(G) < \chi(H)$, evaluation at $k = \chi(G)$ would give $P_G(k) > 0 = P_H(k)$, a contradiction; the reverse inequality is excluded in the same way. No pair claimed in the conjecture exists.

Lean verification and semantic correspondence. The self-contained file `lean/Main.lean` imports only `Std`. Vertices are `Fin n`; the edge relation is arbitrary and decidable, so the result applies in particular to every finite simple graph. The list `words k n` enumerates the length- n lists with entries below k . Theorems `mem_words` and `words_nodup` establish completeness and absence of duplicates. The predicate `Proper` checks the edge constraints, and `chromaticEvaluation` counts the proper assignments. The theorem `evaluation_pos_iff_colorable` proves that this count is positive exactly when a proper coloring exists.

`ChromaticNumberIs` uses the actual existence and minimality definition. The theorem `equal_evaluations_equal_chromatic_numbers` proves equality of chromatic numbers for any two graphs with identical natural coloring counts. Integer polynomials are represented explicitly by coefficient lists and evaluated by Horner's rule in `polynomialEval`. The predicate `IsChromaticPolynomial` is precisely the counting characterization above. The final theorem `not_polynomial_separating_pair_claim` refutes the existence of two graphs sharing an actual integer polynomial with that characterization and having distinct chromatic numbers. Thus the polynomial-to-counting bridge is included in the formal statement, and not left as an assumed numerical coincidence.

Reproduction and trust. Run `lean Main.lean` using Lean 4.19.0. The command exits successfully. The principal results depend only on the standard Lean axioms `propext`, `Classical.choice`, and `Quot.sound`. There are no added axioms, placeholders, or uses of `native_decide`. The proof is uniform in the numbers of vertices and colors and is not limited to a bounded search.